

RESEARCH ASSIGNMENT

Investigate established theory and practical techniques for target acquisition or target identification in noisy sensor



Executive Summary

To reliably track trackside magnet targets using down-facing Hall effect sensors in noisy railway environments, standard radar and sonar signal processing principles can be adapted. High track noise, electromagnetic interference (EMI), mechanical vibration, and non-target ferrous metals create false positives, while high-speed passes or sensor bounce can cause missed detections.

Applying classical radar/sonar principles—specifically **matched filtering, dynamic thresholding, spatial measurement gating, and Bayesian evidence accumulation**—filters out physical noise without risking false counts or missed magnets.

1. Plain-Language Description of the Problem

The core challenge is reliable magnet detection ("target acquisition") under a moving locomotive using magnetic/Hall-effect sensors.

- **The Goal:** Detect fixed, physical magnetic markers (targets) along the track layout to trigger operational events (e.g., speed changes, stop commands, signal triggers) or calibrate dead-reckoning position.
- **The Noise:** The sensor environment is inherently noisy. Sources include wheel/rail magnetic anomalies, track fasteners, electric motor/solenoid EMI, mechanical vibration/chatter causing fluctuating sensor height, duplicate detections from multi-pole magnets or physical bouncing, and electromagnetic spikes during motor state changes.
- **The Failure Modes:**
 - **False Positive (Ghost Target):** Counting a noise spike or nearby iron mass as a magnet, triggering premature automation.
 - **False Negative (Missed Target):** Overshooting or failing to register a real track magnet, causing lost spatial synchronization or overrun.
 - **Duplicate Target:** Registering a single physical magnet multiple times due to field pole transitions or vibration contact chatter.

2 & 3. Target Acquisition Principles & Locomotive Applications

Principle	Technical Source Concept	Application to the Locomotive
1. Matched Filtering / Template Matching	Correlating a known signature against incoming time-series signal to maximize SNR (Signal-to-Noise Ratio) in Gaussian noise.	Correlate incoming Hall voltage against the known characteristic magnetic profile (bell/dipole curve) of the track magnet. Spikes or step-function EMI are rejected.
2. Dynamic Thresholding (CFAR)	Constant False Alarm Rate (CFAR) processing adjusts detection threshold based on local ambient noise floor.	Dynamically raise/lower activation thresholds based on background magnetic noise (e.g., elevated noise when motor current surges).
3. Spatial Measurement Gating	Limiting valid target search space to a "gate" predicted by vehicle kinematics and elapsed time/distance.	If the locomotive travels at V speed, ignore target triggers occurring inside impossible physical distances (e.g., two magnets 1 inch apart if track layout specifies minimum 12 inches).
4. Sequential Evidence Accumulation	Sequential Probability Ratio Test (SPRT) / Bayesian evidence pooling over multiple time samples rather than single-sample thresholding.	Require N consecutive matching samples or integrate magnetic flux over a temporal window matching the physical magnet length at current speed before declaring a valid hit.

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5. Clutter & Duplicate Rejection (Dead-Time / Refractory)	Radar blind-zone / track-management logic to ignore echo or multiple lobe hits from a single target pass.	Enforce a post-detection lockout period (proportional to wheel speed/odometry) to collapse multi-pole transitions into a single validated target pass.
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4 & 5. Assumptions, Risks, & Failure Modes

Matched Filtering

- **Assumptions:** The physical geometry and orientation of track magnets are uniform; velocity is known or bounded so the profile time-width can be calculated.
- **Risks & Failure Modes:** If speed varies drastically during a magnet pass (e.g., rapid acceleration/braking over the magnet), the pulse width stretches or shrinks, degrading filter correlation.

Dynamic Thresholding (CFAR)

- **Assumptions:** Background noise varies slower than the target transit time.
- **Risks & Failure Modes:** A strong continuous magnetic field (e.g., passing over a steel bridge deck or switch frog) can temporarily raise the threshold so high that a genuine weak track magnet is missed.

Kinematic / Spatial Gating

- **Assumptions:** Wheel odometry or speed estimation is reasonably accurate.
- **Risks & Failure Modes:** Wheel slip/skid can corrupt distance estimates, causing the search gate to open early or late and miss a magnet.

Sequential Evidence Accumulation

- **Assumptions:** Sample rate is significantly higher than target transit frequency (Nyquist limit met at maximum train speed).
- **Risks & Failure Modes:** At high speeds, if sample polling rates drop, the magnet transit may yield too few samples to satisfy the accumulation count, causing a missed detection.

6. Unsuitable Techniques

- **High-Order Particle Filtering / Multi-Hypothesis Tracking (MHT):**
 - *Why unsuitable:* Common in radar for tracking dozens of maneuvering targets in 2D/3D space. On a train, targets are 1D, stationary, and fixed to the track layout. Heavy computational overhead offers zero benefit for 1D track-bound geometry.
- **Pure Peak-Detection / Unfiltered Threshold Interrupts:**
 - *Why unsuitable:* Highly susceptible to motor EMI spikes, contact bounce, or metallic debris on the tie. Triggers excessive false positives.
- **Optical / Camera-Based Target Recognition (as sole primary sensor):**
 - *Why unsuitable:* Susceptible to dust, oil, ambient light shifts, and dirt accumulation common in bottom-of-locomotive running gear environments.

7. Smallest Recommended Experiments

Before committing to a production firmware architecture, perform these minimal isolated tests:

1. **Static Baseline & EMI Profile Capture:** Log continuous Hall sensor ADC readings while cycling the locomotive motor through full acceleration, coasting, and hard braking without passing over any magnets. Identify motor frequency spikes and baseline noise floor.
2. **Speed vs. Signature Mapping:** Pass the locomotive over a reference track magnet at three distinct speeds (minimum speed, cruising speed, max speed). Measure pulse duration, signal amplitude variations, and peak sharpness.
3. **Multi-Pole & Bounce Characterization:** Sweep a single sensor over target magnet arrangements to record field reversal signals and measure minimum required software refractory/lockout times.

8. Questions Requiring Operator Judgment

1. **Fail-Safe Orientation:** If a target is missed (due to high speed or extreme noise), should the default action be an emergency stop, a coast to safety, or continued blind navigation on pure odometry?
2. **Layout Minimum Spacing:** What is the absolute minimum physical distance between any two intentional trackside magnets on the layout?
3. **False Positive vs. False Negative Tolerance:** Is it operationally worse to accept a occasional false magnet trigger or to miss a genuine target pass?

Distinctions & Provenance

- **Established Technical Literature:** Matched Filtering (*Skolnik, Radar Handbook*); Constant False Alarm Rate (CFAR) detection (*Richards, Fundamentals of Radar Signal Processing*); Bayesian Sequential Analysis (*Wald, Sequential Analysis*).
- **Railway/On-Train Facts:** Detectors are down-facing Hall effect sensors on a rail vehicle subject to motor EMI, varying speeds, and fixed track-mounted magnets.
- **Inferences:** Dynamic speed compensation is mandatory for matched filtering on rolling stock because pulse duration is inversely proportional to velocity ($t = d/v$).
- **Recommendations:** Implement a lightweight 2-stage processing pipeline in embedded C/C++: Stage 1 = Speed-scaled Matched Filter / Sliding Window Integrator; Stage 2 = Distance-based Spatial Lockout Gate (Refractory Period).